

# APPENDIX

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## LINEAR ALGEBRA

Linear algebra abstracts and generalizes the arithmetic of ordinary vectors, as we encounter them in first-year physics. The generalization is in two directions: (1) we allow the scalars to be *complex* numbers, and (2) we do not restrict ourselves to three dimensions.

### A.1 VECTORS

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A **vector space** consists of a set of **vectors** ( $|\alpha\rangle, |\beta\rangle, |\gamma\rangle, \dots$ ), together with a set of **scalars** ( $a, b, c, \dots$ ),<sup>1</sup> which is **closed**<sup>2</sup> under two operations: vector addition and scalar multiplication.

- **Vector Addition**

The “sum” of any two vectors is another vector:

$$|\alpha\rangle + |\beta\rangle = |\gamma\rangle. \quad (\text{A.1})$$

Vector addition is **commutative**:

$$|\alpha\rangle + |\beta\rangle = |\beta\rangle + |\alpha\rangle, \quad (\text{A.2})$$

and **associative**:

$$|\alpha\rangle + (|\beta\rangle + |\gamma\rangle) = (|\alpha\rangle + |\beta\rangle) + |\gamma\rangle. \quad (\text{A.3})$$

There exists a **zero** (or **null**) **vector**,<sup>3</sup>  $|0\rangle$ , with the property that

$$|\alpha\rangle + |0\rangle = |\alpha\rangle, \quad (\text{A.4})$$

for every vector  $|\alpha\rangle$ . And for every vector  $|\alpha\rangle$  there is an associated **inverse vector** ( $|\alpha\rangle$ ),<sup>4</sup> such that

$$|\alpha\rangle + |-\alpha\rangle = |0\rangle. \quad (\text{A.5})$$

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<sup>1</sup> For our purposes, the scalars will be ordinary complex numbers. Mathematicians can tell you about vector spaces over more exotic fields, but such objects play no role in quantum mechanics. Note that  $\alpha, \beta, \gamma \dots$  are *not* (ordinarily) numbers; they are *names* (labels)—“Charlie,” for instance, or “F43A-9GL,” or whatever you care to use to identify the vector in question.

<sup>2</sup> That is to say, these operations are always well-defined, and will never carry you outside the vector space.

<sup>3</sup> It is customary, where no confusion can arise, to write the null vector without the adorning bracket:  $|0\rangle \rightarrow 0$ .

<sup>4</sup> This is funny notation, since  $\alpha$  is not a number. I’m simply adopting the name “—Charlie” for the inverse of the vector whose name is “Charlie.” More natural terminology will suggest itself in a moment.

• **Scalar Multiplication**

The “product” of any scalar with any vector is another vector:

$$a|\alpha\rangle = |\gamma\rangle. \quad (\text{A.6})$$

Scalar multiplication is **distributive** with respect to vector addition:

$$a(|\alpha\rangle + |\beta\rangle) = a|\alpha\rangle + a|\beta\rangle, \quad (\text{A.7})$$

and with respect to scalar addition:

$$(a + b)|\alpha\rangle = a|\alpha\rangle + b|\alpha\rangle. \quad (\text{A.8})$$

It is also **associative** with respect to the ordinary multiplication of scalars:

$$a(b|\alpha\rangle) = (ab)|\alpha\rangle. \quad (\text{A.9})$$

Multiplication by the scalars 0 and 1 has the effect you would expect:

$$0|\alpha\rangle = |0\rangle; \quad 1|\alpha\rangle = |\alpha\rangle. \quad (\text{A.10})$$

Evidently  $|-1\rangle|\alpha\rangle = (-1)|\alpha\rangle$  (which we write more simply as  $-|\alpha\rangle$ ).

There’s a lot less here than meets the eye—all I have done is to write down in abstract language the familiar rules for manipulating vectors. The virtue of such abstraction is that we will be able to apply our knowledge and intuition about the behavior of ordinary vectors to other systems that happen to share the same formal properties.

A **linear combination** of the vectors  $|\alpha\rangle, |\beta\rangle, |\gamma\rangle, \dots$ , is an expression of the form

$$a|\alpha\rangle + b|\beta\rangle + c|\gamma\rangle + \dots \quad (\text{A.11})$$

A vector  $|\lambda\rangle$  is said to be **linearly independent** of the set  $|\alpha\rangle, |\beta\rangle, |\gamma\rangle, \dots$ , if it cannot be written as a linear combination of them. (For example, in three dimensions the unit vector  $\hat{k}$  is linearly independent of  $\hat{i}$  and  $\hat{j}$ , but any vector in the  $xy$  plane is linearly *dependent* on  $\hat{i}$  and  $\hat{j}$ .) By extension, a *set* of vectors is “linearly independent” if each one is linearly independent of all the rest. A collection of vectors is said to **span** the space if *every* vector can be written as a linear combination of the members of this set.<sup>5</sup> A set of *linearly independent* vectors that spans the space is called a **basis**. The number of vectors in any basis is called the **dimension** of the space. For the moment we shall assume that the dimension ( $n$ ) is *finite*.

With respect to a prescribed basis

$$|e_1\rangle, |e_2\rangle, \dots, |e_n\rangle, \quad (\text{A.12})$$

any given vector

$$|\alpha\rangle = a_1|e_1\rangle + a_2|e_2\rangle + \dots + a_n|e_n\rangle, \quad (\text{A.13})$$

is uniquely represented by the (ordered)  $n$ -tuple of its **components**:

$$|\alpha\rangle \leftrightarrow (a_1, a_2, \dots, a_n). \quad (\text{A.14})$$

<sup>5</sup> A set of vectors that spans the space is also called **complete**, though I personally reserve that word for the infinite-dimensional case, where subtle questions of convergence may arise.

It is often easier to work with the components than with the abstract vectors themselves. To add vectors, you add their corresponding components:

$$|\alpha\rangle + |\beta\rangle \leftrightarrow (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n); \quad (\text{A.15})$$

to multiply by a scalar you multiply each component:

$$c|\alpha\rangle \leftrightarrow (ca_1, ca_2, \dots, ca_n); \quad (\text{A.16})$$

the null vector is represented by a string of zeroes:

$$|0\rangle \leftrightarrow (0, 0, \dots, 0); \quad (\text{A.17})$$

and the components of the inverse vector have their signs reversed:

$$|-\alpha\rangle \leftrightarrow (-a_1, -a_2, \dots, -a_n). \quad (\text{A.18})$$

The only *disadvantage* of working with components is that you have to commit yourself to a particular basis, and the same manipulations will look very different to someone using a different basis.

**Problem A.1** Consider the ordinary vectors in three dimensions  $(a_x\hat{i} + a_y\hat{j} + a_z\hat{k})$ , with complex components.

- (a) Does the subset of all vectors with  $a_z = 0$  constitute a vector space? If so, what is its dimension; if not, why not?
- (b) What about the subset of all vectors whose  $z$  component is 1? *Hint:* Would the sum of two such vectors be in the subset? How about the null vector?
- (c) What about the subset of vectors whose components are all equal?

\* **Problem A.2** Consider the collection of all polynomials (with complex coefficients) of degree  $< N$  in  $x$ .

- (a) Does this set constitute a vector space (with the polynomials as “vectors”)? If so, suggest a convenient basis, and give the dimension of the space. If not, which of the defining properties does it lack?
- (b) What if we require that the polynomials be *even* functions?
- (c) What if we require that the leading coefficient (i.e. the number multiplying  $x^{N-1}$ ) be 1?
- (d) What if we require that the polynomials have the value 0 at  $x = 1$ ?
- (e) What if we require that the polynomials have the value 1 at  $x = 0$ ?

**Problem A.3** Prove that the components of a vector with respect to a given basis are *unique*.

## A.2 INNER PRODUCTS

In three dimensions we encounter two kinds of vector products: the dot product and the cross product. The latter does not generalize in any natural way to  $n$ -dimensional vector spaces, but the former *does*—in this context it is usually called the **inner product**. The inner product

of vectors  $|\alpha\rangle$  and  $|\beta\rangle$  is a complex number (which we write as  $\langle\alpha|\beta\rangle$ ), with the following properties:

$$\langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*, \quad (\text{A.19})$$

$$\langle\alpha|\alpha\rangle \geq 0, \quad \text{and} \quad \langle\alpha|\alpha\rangle = 0 \Leftrightarrow |\alpha\rangle = |0\rangle, \quad (\text{A.20})$$

$$\langle\alpha|(b|\beta\rangle + c|\gamma\rangle) = b\langle\alpha|\beta\rangle + c\langle\alpha|\gamma\rangle. \quad (\text{A.21})$$

Apart from the generalization to complex numbers, these axioms simply codify the familiar behavior of dot products. A vector space with an inner product is called an **inner product space**.

Because the inner product of any vector with itself is a non-negative number (Equation A.20), its square root is *real*—we call this the **norm** of the vector:

$$\|\alpha\| \equiv \sqrt{\langle\alpha|\alpha\rangle}; \quad (\text{A.22})$$

it generalizes the notion of “length.” A **unit vector** (one whose norm is 1) is said to be **normalized** (the word should really be “normal,” but I guess that sounds too judgmental). Two vectors whose inner product is zero are called **orthogonal** (generalizing the notion of “perpendicular”). A collection of mutually orthogonal normalized vectors,

$$\langle\alpha_i|\alpha_j\rangle = \delta_{ij}, \quad (\text{A.23})$$

is called an **orthonormal set**. It is always possible (see Problem A.4), and almost always convenient, to choose an **orthonormal basis**; in that case the inner product of two vectors can be written very neatly in terms of their components:

$$\langle\alpha|\beta\rangle = a_1^*b_1 + a_2^*b_2 + \cdots + a_n^*b_n, \quad (\text{A.24})$$

the norm (squared) becomes

$$\langle\alpha|\alpha\rangle = |a_1|^2 + |a_2|^2 + \cdots + |a_n|^2, \quad (\text{A.25})$$

and the components themselves are

$$a_i = \langle e_i|\alpha\rangle. \quad (\text{A.26})$$

(These results generalize the familiar formulas  $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$ ,  $|\mathbf{a}|^2 = a_x^2 + a_y^2 + a_z^2$ , and  $a_x = \hat{i} \cdot \mathbf{a}$ ,  $a_y = \hat{j} \cdot \mathbf{a}$ ,  $a_z = \hat{k} \cdot \mathbf{a}$ , for the three-dimensional orthonormal basis  $\hat{i}$ ,  $\hat{j}$ ,  $\hat{k}$ .) From now on we shall *always* work in orthonormal bases, unless it is explicitly indicated otherwise.

Another geometrical quantity one might wish to generalize is the *angle* between two vectors. In ordinary vector analysis  $\cos \theta = (\mathbf{a} \cdot \mathbf{b}) / |\mathbf{a}| |\mathbf{b}|$ , but because the inner product is in general a complex number, the analogous formula (in an arbitrary inner product space) does not define a (real) angle  $\theta$ . Nevertheless, it is still true that the *absolute value* of this quantity is a number no greater than 1,

$$|\langle\alpha|\beta\rangle|^2 \leq \langle\alpha|\alpha\rangle \langle\beta|\beta\rangle. \quad (\text{A.27})$$

This important result is known as the **Schwarz inequality**; the proof is given in Problem A.5. So you can, if you like, define the angle between  $|\alpha\rangle$  and  $|\beta\rangle$  by the formula

$$\cos \theta = \sqrt{\frac{\langle\alpha|\beta\rangle \langle\beta|\alpha\rangle}{\langle\alpha|\alpha\rangle \langle\beta|\beta\rangle}}. \quad (\text{A.28})$$

\* **Problem A.4** Suppose you start out with a basis  $(|e_1\rangle, |e_2\rangle, \dots, |e_n\rangle)$  that is *not* orthonormal. The **Gram–Schmidt procedure** is a systematic ritual for generating from it an orthonormal basis  $(|e'_1\rangle, |e'_2\rangle, \dots, |e'_n\rangle)$ . It goes like this:

(i) Normalize the first basis vector (divide by its norm):

$$|e'_1\rangle = \frac{|e_1\rangle}{\|e_1\|}.$$

(ii) Find the projection of the second vector along the first, and subtract it off:

$$|e_2\rangle - \langle e'_1 | e_2 \rangle |e'_1\rangle.$$

This vector is orthogonal to  $|e'_1\rangle$ ; normalize it to get  $|e'_2\rangle$ .

(iii) Subtract from  $|e_3\rangle$  its projections along  $|e'_1\rangle$  and  $|e'_2\rangle$ :

$$|e_3\rangle - \langle e'_1 | e_3 \rangle |e'_1\rangle - \langle e'_2 | e_3 \rangle |e'_2\rangle.$$

This is orthogonal to  $|e'_1\rangle$  and  $|e'_2\rangle$ ; normalize it to get  $|e'_3\rangle$ . And so on.

Use the Gram–Schmidt procedure to orthonormalize the 3-space basis  $|e_1\rangle = (1 + i)\hat{i} + (1)\hat{j} + (i)\hat{k}$ ,  $|e_2\rangle = (i)\hat{i} + (3)\hat{j} + (1)\hat{k}$ ,  $|e_3\rangle = (0)\hat{i} + (28)\hat{j} + (0)\hat{k}$ .

**Problem A.5** Prove the Schwarz inequality (Equation A.27). *Hint:* Let  $|\gamma\rangle = |\beta\rangle - (\langle\alpha|\beta\rangle / \langle\alpha|\alpha\rangle) |\alpha\rangle$ , and use  $\langle\gamma|\gamma\rangle \geq 0$ .

**Problem A.6** Find the angle (in the sense of Equation A.28) between the vectors  $|\alpha\rangle = (1 + i)\hat{i} + (1)\hat{j} + (i)\hat{k}$  and  $|\beta\rangle = (4 - i)\hat{i} + (0)\hat{j} + (2 - 2i)\hat{k}$ .

**Problem A.7** Prove the **triangle inequality**:  $\|(|\alpha\rangle + |\beta\rangle)\| \leq \|\alpha\| + \|\beta\|$ .

### A.3 MATRICES

Suppose you take every vector (in 3-space) and multiply it by 17, or you rotate every vector by  $39^\circ$  about the  $z$  axis, or you reflect every vector in the  $xy$  plane—these are all examples of **linear transformations**. A linear transformation<sup>6</sup> ( $\hat{T}$ ) takes each vector in a vector space and “transforms” it into some other vector ( $|\alpha\rangle \rightarrow |\alpha'\rangle = \hat{T}|\alpha\rangle$ ), subject to the condition that the operation be *linear*:

$$\hat{T}(a|\alpha\rangle + b|\beta\rangle) = a(\hat{T}|\alpha\rangle) + b(\hat{T}|\beta\rangle), \quad (\text{A.29})$$

for any vectors  $|\alpha\rangle, |\beta\rangle$  and any scalars  $a, b$ .

If you know what a particular linear transformation does to a set of *basis* vectors, you can easily figure out what it does to *any* vector. For suppose that

<sup>6</sup> In this chapter I’ll use a hat (^) to denote linear transformations; this is not inconsistent with my convention in the text (putting hats on operators), for (as we shall see) quantum operators *are* linear transformations.

$$\begin{aligned}
\hat{T}|e_1\rangle &= T_{11}|e_1\rangle + T_{21}|e_2\rangle + \cdots + T_{n1}|e_n\rangle, \\
\hat{T}|e_2\rangle &= T_{12}|e_1\rangle + T_{22}|e_2\rangle + \cdots + T_{n2}|e_n\rangle, \\
&\vdots \\
\hat{T}|e_n\rangle &= T_{1n}|e_1\rangle + T_{2n}|e_2\rangle + \cdots + T_{nn}|e_n\rangle,
\end{aligned}$$

or, more compactly,

$$\hat{T}|e_j\rangle = \sum_{i=1}^n T_{ij}|e_i\rangle, \quad (j = 1, 2, \dots, n). \quad (\text{A.30})$$

If  $|\alpha\rangle$  is an arbitrary vector,

$$|\alpha\rangle = a_1|e_1\rangle + a_2|e_2\rangle + \cdots + a_n|e_n\rangle = \sum_{j=1}^n a_j|e_j\rangle, \quad (\text{A.31})$$

then

$$\hat{T}|\alpha\rangle = \sum_{j=1}^n a_j (\hat{T}|e_j\rangle) = \sum_{j=1}^n \sum_{i=1}^n a_j T_{ij}|e_i\rangle = \sum_{i=1}^n \left( \sum_{j=1}^n T_{ij} a_j \right) |e_i\rangle. \quad (\text{A.32})$$

Evidently  $\hat{T}$  takes a vector with components  $a_1, a_2, \dots, a_n$  into a vector with components<sup>7</sup>

$$a'_i = \sum_{j=1}^n T_{ij} a_j. \quad (\text{A.33})$$

Thus the  $n^2$  **elements**  $T_{ij}$  uniquely characterize the linear transformation  $\hat{T}$  (with respect to a given basis), just as the  $n$  components  $a_i$  uniquely characterize the vector  $|\alpha\rangle$  (with respect to that basis):

$$\hat{T} \leftrightarrow (T_{11}, T_{12}, \dots, T_{nn}). \quad (\text{A.34})$$

If the basis is orthonormal, it follows from Equation A.30 that

$$T_{ij} = \langle e_i | \hat{T} | e_j \rangle. \quad (\text{A.35})$$

It is convenient to display these complex numbers in the form of a **matrix**:<sup>8</sup>

$$\mathbf{T} = \begin{pmatrix} T_{11} & T_{12} & \cdots & T_{1n} \\ T_{21} & T_{22} & \cdots & T_{2n} \\ \vdots & \vdots & & \vdots \\ T_{n1} & T_{n2} & \cdots & T_{nn} \end{pmatrix}. \quad (\text{A.36})$$

The study of linear transformations reduces, then, to the theory of matrices. The *sum* of two linear transformations  $(\hat{S} + \hat{T})$  is defined in the natural way:

$$(\hat{S} + \hat{T})|\alpha\rangle = \hat{S}|\alpha\rangle + \hat{T}|\alpha\rangle; \quad (\text{A.37})$$

<sup>7</sup> Notice the reversal of indices between Equations A.30 and A.33. This is not a typographical error. Another way of putting it (switching  $i \leftrightarrow j$  in Equation A.30) is that if the *components* transform with  $T_{ij}$ , the *basis* vectors transform with  $T_{ji}$ .

<sup>8</sup> I'll use boldface capital letters, sans serif, to denote square matrices.

this matches the usual rule for adding matrices (you add the corresponding elements):

$$\mathbf{U} = \mathbf{S} + \mathbf{T} \Leftrightarrow U_{ij} = S_{ij} + T_{ij}. \quad (\text{A.38})$$

The *product* of two linear transformations  $(\hat{S}\hat{T})$  is the net effect of performing them in succession—first  $\hat{T}$ , then  $\hat{S}$ :

$$|\alpha'\rangle = \hat{T}|\alpha\rangle; \quad |\alpha''\rangle = \hat{S}|\alpha'\rangle = \hat{S}(\hat{T}|\alpha\rangle) = \hat{S}\hat{T}|\alpha\rangle. \quad (\text{A.39})$$

What matrix  $\mathbf{U}$  represents the combined transformation  $\hat{U} = \hat{S}\hat{T}$ ? It's not hard to work it out:

$$a_i'' = \sum_{j=1}^n S_{ij} a_j' = \sum_{j=1}^n S_{ij} \left( \sum_{k=1}^n T_{jk} a_k \right) = \sum_{k=1}^n \left( \sum_{j=1}^n S_{ij} T_{jk} \right) a_k = \sum_{k=1}^n U_{ik} a_k.$$

Evidently

$$\mathbf{U} = \mathbf{S}\mathbf{T} \Leftrightarrow U_{ik} = \sum_{j=1}^n S_{ij} T_{jk} \quad (\text{A.40})$$

—this is the standard rule for matrix multiplication: to find the  $ik$ th element of the product, you look at the  $i$ th row of  $\mathbf{S}$ , and the  $k$ th column of  $\mathbf{T}$ , multiply corresponding entries, and add. The same prescription allows you to multiply *rectangular* matrices, as long as the number of columns in the first matches the number of rows in the second. In particular, if we write the  $n$ -tuple of components of  $|\alpha\rangle$  as an  $n \times 1$  **column matrix** (or “column vector”):<sup>9</sup>

$$\mathbf{a} \equiv \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \quad (\text{A.41})$$

the transformation rule (Equation A.33) can be expressed as a matrix product:

$$\mathbf{a}' = \mathbf{T}\mathbf{a}. \quad (\text{A.42})$$

Now some matrix terminology:

- The **transpose** of a matrix (which we shall write with a tilde:  $\tilde{\mathbf{T}}$ ) is the same set of elements, but with rows and columns interchanged. In particular, the transpose of a *column* matrix is a **row matrix**:

$$\tilde{\mathbf{a}} = (a_1 \quad a_2 \quad \dots \quad a_n). \quad (\text{A.43})$$

For a *square* matrix taking the transpose amounts to reflecting in the **main diagonal** (upper left to lower right):

$$\tilde{\mathbf{T}} = \begin{pmatrix} T_{11} & T_{21} & \dots & T_{n1} \\ T_{12} & T_{22} & \dots & T_{n2} \\ \vdots & \vdots & & \vdots \\ T_{1n} & T_{2n} & \dots & T_{nn} \end{pmatrix}. \quad (\text{A.44})$$

<sup>9</sup> I'll use a boldface lower-case letters, sans serif, for row and column matrices.

A (square) matrix is **symmetric** if it is equal to its transpose; it is **antisymmetric** if this operation reverses the sign:

$$\text{symmetric : } \tilde{T} = T; \quad \text{antisymmetric : } \tilde{T} = -T. \quad (\text{A.45})$$

- The (complex) **conjugate** of a matrix (which we denote, as usual, with an asterisk,  $T^*$ ), consists of the complex conjugate of every element:

$$T^* = \begin{pmatrix} T_{11}^* & T_{12}^* & \cdots & T_{1n}^* \\ T_{21}^* & T_{22}^* & \cdots & T_{2n}^* \\ \vdots & \vdots & & \vdots \\ T_{n1}^* & T_{n2}^* & \cdots & T_{nn}^* \end{pmatrix}; \quad a^* = \begin{pmatrix} a_1^* \\ a_2^* \\ \vdots \\ a_n^* \end{pmatrix}. \quad (\text{A.46})$$

A matrix is **real** if all its elements are real, and **imaginary** if they are all imaginary:

$$\text{real : } T^* = T; \quad \text{imaginary : } T^* = -T. \quad (\text{A.47})$$

- The **hermitian conjugate** (or **adjoint**) of a matrix (indicated by a dagger,  $T^\dagger$ ) is the transpose conjugate:

$$T^\dagger \equiv \tilde{T}^* = \begin{pmatrix} T_{11}^* & T_{21}^* & \cdots & T_{n1}^* \\ T_{12}^* & T_{22}^* & \cdots & T_{n2}^* \\ \vdots & \vdots & & \vdots \\ T_{1n}^* & T_{2n}^* & \cdots & T_{nn}^* \end{pmatrix}; \quad a^\dagger \equiv \tilde{a}^* = (a_1^* \ a_2^* \ \cdots \ a_n^*). \quad (\text{A.48})$$

A square matrix is **hermitian** (or **self-adjoint**) if it is equal to its hermitian conjugate; if hermitian conjugation introduces a minus sign, the matrix is **skew hermitian** (or **anti-hermitian**):

$$\text{hermitian : } T^\dagger = T; \quad \text{skew hermitian : } T^\dagger = -T. \quad (\text{A.49})$$

In this notation the inner product of two vectors (with respect to an orthonormal basis—Equation A.24), can be written very neatly as a matrix product:

$$\langle \alpha | \beta \rangle = a^\dagger b. \quad (\text{A.50})$$

Notice that each of the three operations defined in this paragraph, if applied twice, returns you to the original matrix.

Matrix multiplication is not, in general, commutative ( $ST \neq TS$ ); the *difference* between the two orderings is called the **commutator**:<sup>10</sup>

$$[S, T] \equiv ST - TS. \quad (\text{A.51})$$

The transpose of a product is the product of the transposes *in reverse order*:

$$(\tilde{ST}) = \tilde{T}\tilde{S}, \quad (\text{A.52})$$

(see Problem A.11), and the same goes for hermitian conjugates:

$$(ST)^\dagger = T^\dagger S^\dagger. \quad (\text{A.53})$$

<sup>10</sup> The commutator only makes sense for *square* matrices, of course; for rectangular matrices the two orderings wouldn't even be the same size.



The **identity matrix** (representing a linear transformation that carries every vector into itself) consists of ones on the main diagonal, and zeroes everywhere else:

$$I \equiv \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}. \quad (\text{A.54})$$

In other words,

$$I_{ij} = \delta_{ij}. \quad (\text{A.55})$$

The **inverse** of a (square) matrix (written  $T^{-1}$ ) is defined in the obvious way:<sup>11</sup>

$$T^{-1}T = TT^{-1} = I. \quad (\text{A.56})$$

A matrix has an inverse if and only if its **determinant**<sup>12</sup> is nonzero; in fact,

$$T^{-1} = \frac{1}{\det T} \tilde{C}, \quad (\text{A.57})$$

where  $\tilde{C}$  is the matrix of **cofactors** (the cofactor of element  $T_{ij}$  is  $(-1)^{i+j}$  times the determinant of the submatrix obtained from  $T$  by erasing the  $i$ th row and the  $j$ th column). A matrix that has no inverse is said to be **singular**. The inverse of a product (assuming it exists) is the product of the inverses *in reverse order*:

$$(ST)^{-1} = T^{-1}S^{-1}. \quad (\text{A.58})$$

A matrix is **unitary** if its inverse is equal to its hermitian conjugate.<sup>13</sup>

$$\text{unitary} : U^\dagger = U^{-1}. \quad (\text{A.59})$$

Assuming the basis is orthonormal, the columns of a unitary matrix constitute an orthonormal set, and so too do its rows (see Problem A.12). Linear transformations represented by unitary matrices preserve inner products, since (Equation A.50)

$$\langle \alpha' | \beta' \rangle = a'^\dagger b' = (Ua)^\dagger (Ub) = a^\dagger U^\dagger U b = a^\dagger b = \langle \alpha | \beta \rangle. \quad (\text{A.60})$$

\* **Problem A.8** Given the following two matrices:

$$A = \begin{pmatrix} -1 & 1 & i \\ 2 & 0 & 3 \\ 2i & -2i & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & -i \\ 0 & 1 & 0 \\ i & 3 & 2 \end{pmatrix},$$

compute: (a)  $A + B$ , (b)  $AB$ , (c)  $[A, B]$ , (d)  $\tilde{A}$ , (e)  $A^*$ , (f)  $A^\dagger$ , (g)  $\det(B)$ , and (h)  $B^{-1}$ . Check that  $B B^{-1} = I$ . Does  $A$  have an inverse?

<sup>11</sup> Note that the left inverse is equal to the right inverse, for if  $AT = I$  and  $TB = I$ , then (multiplying the second on the left by  $A$  and invoking the first) we get  $B = A$ .

<sup>12</sup> I assume you know how to evaluate determinants. If not, see Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd edn (John Wiley, New York, 2006), Section 3.3.

<sup>13</sup> In a *real* vector space (that is, one in which the scalars are real) the hermitian conjugate is the same as the transpose, and a unitary matrix is **orthogonal**:  $\tilde{O} = O^{-1}$ . For example, rotations in ordinary 3-space are represented by orthogonal matrices.

\* **Problem A.9** Using the square matrices in Problem A.8, and the column matrices

$$\mathbf{a} = \begin{pmatrix} i \\ 2i \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ (1-i) \\ 0 \end{pmatrix},$$

find: (a)  $\mathbf{Aa}$ , (b)  $\mathbf{a}^\dagger \mathbf{b}$ , (c)  $\tilde{\mathbf{a}} \mathbf{B} \mathbf{b}$ , (d)  $\mathbf{a} \mathbf{b}^\dagger$ .

**Problem A.10** By explicit construction of the matrices in question, show that any matrix  $\mathbf{T}$  can be written

- (a) as the sum of a symmetric matrix  $\mathbf{S}$  and an antisymmetric matrix  $\mathbf{A}$ ;
- (b) as the sum of a real matrix  $\mathbf{R}$  and an imaginary matrix  $\mathbf{M}$ ;
- (c) as the sum of a hermitian matrix  $\mathbf{H}$  and a skew-hermitian matrix  $\mathbf{K}$ .

\* **Problem A.11** Prove Equations A.52, A.53, and A.58. Show that the product of two unitary matrices is unitary. Under what conditions is the product of two hermitian matrices hermitian? Is the sum of two unitary matrices necessarily unitary? Is the sum of two hermitian matrices always hermitian?

**Problem A.12** Show that the rows and columns of a unitary matrix constitute orthonormal sets.

**Problem A.13** Noting that  $\det(\tilde{\mathbf{T}}) = \det(\mathbf{T})$ , show that the determinant of a hermitian matrix is real, the determinant of a unitary matrix has modulus 1 (hence the name), and the determinant of an orthogonal matrix (footnote 13) is either +1 or -1.

## A.4 CHANGING BASES

The components of a vector depend, of course, on your (arbitrary) choice of basis, and so do the elements of the matrix representing a linear transformation. We might inquire how these numbers change when we switch to a different basis. The old basis vectors,  $|e_i\rangle$  are—like *all* vectors—linear combinations of the new ones,  $|f_i\rangle$ :

$$\begin{aligned} |e_1\rangle &= S_{11}|f_1\rangle + S_{21}|f_2\rangle + \cdots + S_{n1}|f_n\rangle, \\ |e_2\rangle &= S_{12}|f_1\rangle + S_{22}|f_2\rangle + \cdots + S_{n2}|f_n\rangle, \\ &\vdots \\ |e_n\rangle &= S_{1n}|f_1\rangle + S_{2n}|f_2\rangle + \cdots + S_{nn}|f_n\rangle, \end{aligned}$$

(for some set of complex numbers  $S_{ij}$ ), or, more compactly,

$$|e_j\rangle = \sum_{i=1}^n S_{ij}|f_i\rangle, \quad (j = 1, 2, \dots, n). \quad (\text{A.61})$$

This is *itself* a linear transformation (compare Equation A.30),<sup>14</sup> and we know immediately how the components transform:

$$a_i^f = \sum_{j=1}^n S_{ij}a_j^e, \quad (\text{A.62})$$

(where the superscript indicates the basis). In matrix form

$$\mathbf{a}^f = \mathbf{S}\mathbf{a}^e. \quad (\text{A.63})$$

What about the matrix representing a linear transformation  $\hat{T}$ —how is *it* modified by a change of basis? Well, in the old basis we had (Equation A.42)

$$\mathbf{a}'^e = \mathbf{T}^e\mathbf{a}^e,$$

and Equation A.63—multiplying both sides by  $\mathbf{S}^{-1}$ —entails<sup>15</sup>  $\mathbf{a}^e = \mathbf{S}^{-1}\mathbf{a}^f$ , so

$$\mathbf{a}'^f = \mathbf{S}\mathbf{a}'^e = \mathbf{S}(\mathbf{T}^e\mathbf{a}^e) = \mathbf{S}\mathbf{T}^e\mathbf{S}^{-1}\mathbf{a}^f.$$

Evidently

$$\mathbf{T}^f = \mathbf{S}\mathbf{T}^e\mathbf{S}^{-1}. \quad (\text{A.64})$$

In general, two matrices ( $\mathbf{T}_1$  and  $\mathbf{T}_2$ ) are said to be **similar** if  $\mathbf{T}_2 = \mathbf{S}\mathbf{T}_1\mathbf{S}^{-1}$  for some (non-singular) matrix  $\mathbf{S}$ . What we have just found is that *matrices representing the same linear transformation, with respect to different bases, are similar*. Incidentally, if the first basis is orthonormal, the second will also be orthonormal if and only if the matrix  $\mathbf{S}$  is *unitary* (see Problem A.16). Since we always work in orthonormal bases, we are interested mainly in *unitary* similarity transformations.

While the *elements* of the matrix representing a given linear transformation may look very different in the new basis, *two* numbers associated with the matrix are unchanged: the determinant and the **trace**. For the determinant of a product is the product of the determinants, and hence

$$\det(\mathbf{T}^f) = \det(\mathbf{S}\mathbf{T}^e\mathbf{S}^{-1}) = \det(\mathbf{S})\det(\mathbf{T}^e)\det(\mathbf{S}^{-1}) = \det\mathbf{T}^e. \quad (\text{A.65})$$

And the trace, which is the *sum of the diagonal elements*,

$$\text{Tr}(\mathbf{T}) \equiv \sum_{i=1}^m T_{ii}, \quad (\text{A.66})$$

has the property (see Problem A.17) that

$$\text{Tr}(\mathbf{T}_1\mathbf{T}_2) = \text{Tr}(\mathbf{T}_2\mathbf{T}_1), \quad (\text{A.67})$$

<sup>14</sup> Notice, however, the radically different perspective: In this case we're talking about one and the same *vector*, referred to two completely different *bases*, whereas before we were thinking of a completely *different* vector, referred to the *same* basis.

<sup>15</sup> Note that  $\mathbf{S}^{-1}$  certainly exists—if  $\mathbf{S}$  were singular, the  $|f_i\rangle$ s would not span the space, so they wouldn't constitute a basis.

(for any two matrices  $T_1$  and  $T_2$ ), so

$$\text{Tr}(T^f) = \text{Tr}(ST^eS^{-1}) = \text{Tr}(T^eS^{-1}S) = \text{Tr}(T^e). \quad (\text{A.68})$$

**Problem A.14** Using the standard basis  $(\hat{i}, \hat{j}, \hat{k})$  for vectors in three dimensions:

- (a) Construct the matrix representing a rotation through angle  $\theta$  (counterclockwise, looking down the axis toward the origin) about the  $z$  axis.
- (b) Construct the matrix representing a rotation by  $120^\circ$  (counterclockwise, looking down the axis) about an axis through the point  $(1,1,1)$ .
- (c) Construct the matrix representing reflection through the  $xy$  plane.
- (d) Check that all these matrices are orthogonal, and calculate their determinants.

**Problem A.15** In the usual basis  $(\hat{i}, \hat{j}, \hat{k})$ , construct the matrix  $T_x$  representing a rotation through angle  $\theta$  about the  $x$  axis, and the matrix  $T_y$  representing a rotation through angle  $\theta$  about the  $y$  axis. Suppose now we change bases, to  $\hat{i}' = \hat{j}$ ,  $\hat{j}' = -\hat{i}$ ,  $\hat{k}' = \hat{k}$ . Construct the matrix  $S$  that effects this change of basis, and check that  $ST_xS^{-1}$  and  $ST_yS^{-1}$  are what you would expect.

**Problem A.16** Show that similarity preserves matrix multiplication (that is, if  $A^e B^e = C^e$ , then  $A^f B^f = C^f$ ). Similarity does *not*, in general, preserve symmetry, reality, or hermiticity; show, however, that if  $S$  is *unitary*, and  $H^e$  is hermitian, then  $H^f$  is hermitian. Show that  $S$  carries an orthonormal basis into another orthonormal basis if and only if it is unitary.

\* **Problem A.17** Prove that  $\text{Tr}(T_1 T_2) = \text{Tr}(T_2 T_1)$ . It follows immediately that  $\text{Tr}(T_1 T_2 T_3) = \text{Tr}(T_2 T_3 T_1)$ , but is it the case that  $\text{Tr}(T_1 T_2 T_3) = \text{Tr}(T_2 T_1 T_3)$ , in general? Prove it, or disprove it. *Hint:* The best disproof is always a counterexample—the simpler the better!

## A.5 EIGENVECTORS AND EIGENVALUES

Consider the linear transformation in 3-space consisting of a rotation, about some specified axis, by an angle  $\theta$ . Most vectors (with tails at the origin) will change in a rather complicated way (they ride around on a cone about the axis), but vectors that happen to lie *along* the axis have very simple behavior: They don't change at all ( $\hat{T}|\alpha\rangle = |\alpha\rangle$ ). If  $\theta$  is  $180^\circ$ , then vectors which lie in the “equatorial” plane reverse signs ( $\hat{T}|\alpha\rangle = -|\alpha\rangle$ ). In a complex vector space<sup>16</sup> every linear transformation has “special” vectors like these, which are transformed into scalar multiples of themselves:

<sup>16</sup> This is *not* always true in a *real* vector space (where the scalars are restricted to real values). See Problem A.18.

$$\hat{T}|\alpha\rangle = \lambda|\alpha\rangle; \quad (\text{A.69})$$

they are called **eigenvectors** of the transformation, and the (complex) number  $\lambda$  is their **eigenvalue**. (The *null* vector doesn't count, even though in a trivial sense it obeys Equation A.69 for *any*  $\hat{T}$  and *any*  $\lambda$ ; technically, an eigenvector is any *nonzero* vector satisfying Equation A.69.) Notice that any (nonzero) *multiple* of an eigenvector is still an eigenvector, with the same eigenvalue.

With respect to a particular basis, the eigenvector equation assumes the matrix form

$$T\mathbf{a} = \lambda\mathbf{a}, \quad (\mathbf{a} \neq 0), \quad (\text{A.70})$$

or

$$(T - \lambda I)\mathbf{a} = 0. \quad (\text{A.71})$$

(Here  $0$  is the (column) matrix whose elements are all zero.) Now, if the matrix  $(T - \lambda I)$  had an *inverse*, we could multiply both sides of Equation A.71 by  $(T - \lambda I)^{-1}$ , and conclude that  $\mathbf{a} = 0$ . But by assumption  $\mathbf{a}$  is *not* zero, so the matrix  $(T - \lambda I)$  must in fact be singular, which means that its determinant is zero:

$$\det(T - \lambda I) = \begin{vmatrix} (T_{11} - \lambda) & T_{12} & \cdots & T_{1n} \\ T_{21} & (T_{22} - \lambda) & \cdots & T_{2n} \\ \vdots & \vdots & & \vdots \\ T_{n1} & T_{n2} & \cdots & (T_{nn} - \lambda) \end{vmatrix} = 0. \quad (\text{A.72})$$

Expansion of the determinant yields an algebraic equation for  $\lambda$ :

$$C_n\lambda^n + C_{n-1}\lambda^{n-1} + \cdots + C_1\lambda + C_0 = 0, \quad (\text{A.73})$$

where the coefficients  $C_i$  depend on the elements of  $T$  (see Problem A.20). This is called the **characteristic equation** for the matrix; its solutions determine the eigenvalues. Notice that it's an  $n$ th-order equation, so (by the **fundamental theorem of algebra**) it has  $n$  (complex) roots.<sup>17</sup> However, some of these may be multiple roots, so all we can say for certain is that an  $n \times n$  matrix has *at least one* and *at most*  $n$  distinct eigenvalues. The collection of all the eigenvalues of a matrix is called its **spectrum**; if two (or more) linearly independent eigenvectors share the same eigenvalue, the spectrum is said to be **degenerate**.

To construct the *eigenvectors* it is generally easiest simply to plug each  $\lambda$  back into Equation A.70 and solve “by hand” for the components of  $\mathbf{a}$ . I'll show you how it goes by working out an example.

### Example A.1

Find the eigenvalues and eigenvectors of the following matrix:

$$M = \begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix}. \quad (\text{A.74})$$

<sup>17</sup> It is here that the case of *real* vector spaces becomes more awkward, because the characteristic equation need not have any (real) solutions at all. See Problem A.18.

**Solution:** The characteristic equation is

$$\begin{vmatrix} (2 - \lambda) & 0 & -2 \\ -2i & (i - \lambda) & 2i \\ 1 & 0 & (-1 - \lambda) \end{vmatrix} = -\lambda^3 + (1 + i)\lambda^2 - i\lambda = 0, \quad (\text{A.75})$$

and its roots are 0, 1, and  $i$ . Call the components of the first eigenvector  $(a_1, a_2, a_3)$ ; then

$$\begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = 0 \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

which yields three equations:

$$\begin{aligned} 2a_1 - 2a_3 &= 0, \\ -2ia_1 + ia_2 + 2ia_3 &= 0, \\ a_1 - a_3 &= 0. \end{aligned}$$

The first determines  $a_3$  (in terms of  $a_1$ ):  $a_3 = a_1$ ; the second determines  $a_2$ :  $a_2 = 0$ ; and the third is redundant. We may as well pick  $a_1 = 1$  (since any multiple of an eigenvector is still an eigenvector):

$$\mathbf{a}^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \text{ for } \lambda_1 = 0. \quad (\text{A.76})$$

For the second eigenvector (recycling the same notation for the components) we have

$$\begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = 1 \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix},$$

which leads to the equations

$$\begin{aligned} 2a_1 - 2a_3 &= a_1, \\ -2ia_1 + ia_2 + 2ia_3 &= a_2, \\ a_1 - a_3 &= a_3, \end{aligned}$$

with the solution  $a_3 = (1/2)a_1$ ,  $a_2 = [(1 - i)/2]a_1$ ; this time I'll pick  $a_1 = 2$ , so

$$\mathbf{a}^{(2)} = \begin{pmatrix} 2 \\ 1 - i \\ 1 \end{pmatrix}, \text{ for } \lambda_2 = 1. \quad (\text{A.77})$$

Finally, for the third eigenvector,

$$\begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = i \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} ia_1 \\ ia_2 \\ ia_3 \end{pmatrix},$$

which gives the equations

$$\begin{aligned} 2a_1 - 2a_3 &= ia_1, \\ -2ia_1 + ia_2 + 2ia_3 &= ia_2, \\ a_1 - a_3 &= ia_3, \end{aligned}$$

whose solution is  $a_3 = a_1 = 0$ , with  $a_2$  undetermined. Choosing  $a_2 = 1$ , we conclude

$$\mathbf{a}^{(3)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \text{ for } \lambda_3 = i. \quad (\text{A.78})$$

If the eigenvectors span the space (as they do in the preceding example), we are free to use *them* as a basis:

$$\hat{T}|f_1\rangle = \lambda_1|f_1\rangle,$$

$$\hat{T}|f_2\rangle = \lambda_2|f_2\rangle,$$

...

$$\hat{T}|f_n\rangle = \lambda_n|f_n\rangle.$$

In this basis the matrix representing  $\hat{T}$  takes on a very simple form, with the eigenvalues strung out along the main diagonal, and all other elements zero:

$$\mathbf{T} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}, \quad (\text{A.79})$$

and the (normalized) eigenvectors are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}. \quad (\text{A.80})$$

A matrix that can be brought to **diagonal form** (Equation A.79) by a change of basis is said to be **diagonalizable** (evidently a matrix is diagonalizable if and only if its eigenvectors span the space). The similarity matrix that effects the diagonalization can be constructed by using the eigenvectors (in the old basis) as the columns of  $\mathbf{S}^{-1}$ :

$$(\mathbf{S}^{-1})_{ij} = (\mathbf{a}^{(j)})_i. \quad (\text{A.81})$$

### Example A.2

In Example A.1,

$$\mathbf{S}^{-1} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & (1-i) & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

so (using Equation A.57)

$$\mathbf{S} = \begin{pmatrix} -1 & 0 & 2 \\ 1 & 0 & -1 \\ (i-1) & 1 & (1-i) \end{pmatrix};$$

you can check for yourself that

$$S\mathbf{a}^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad S\mathbf{a}^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad S\mathbf{a}^{(3)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

and

$$SMS^{-1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & i \end{pmatrix}.$$

There's an obvious advantage in bringing a matrix to diagonal form: it's much easier to work with. Unfortunately, not every matrix *can* be diagonalized—the eigenvectors have to span the space. If the characteristic equation has  $n$  distinct roots, then the matrix is certainly diagonalizable, but it *may* be diagonalizable even if there are multiple roots. (For an example of a matrix that *cannot* be diagonalized, see Problem A.19.) It would be handy to know in advance (before working out all the eigenvectors) whether a given matrix is diagonalizable. A useful sufficient (though not necessary) condition is the following: A matrix is said to be **normal** if it commutes with its hermitian conjugate:

$$\text{normal : } [\mathbf{N}^\dagger, \mathbf{N}] = 0. \quad (\text{A.82})$$

Every normal matrix is diagonalizable (its eigenvectors span the space). In particular, every hermitian matrix is diagonalizable, and so is every unitary matrix.

Suppose we have *two* diagonalizable matrices; in quantum applications the question often arises: Can they be **simultaneously diagonalized** (by the *same* similarity matrix  $S$ )? That is to say, does there exist a basis all of whose members are eigenvectors of *both* matrices? In this basis, both matrices would be diagonal. The answer is yes *if and only if the two matrices commute*, as we shall now prove. (By the way, if two matrices commute with respect to *one* basis, they commute with respect to *any* basis—see Problem A.23.)

We first show that if a basis of simultaneous eigenvectors exists then the matrices commute. Actually, it's trivial in the (simultaneously) diagonal form:

$$\begin{aligned} TV &= \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} v_1 & 0 & \cdots & 0 \\ 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_n \end{pmatrix} \\ &= \begin{pmatrix} \lambda_1 v_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n v_n \end{pmatrix} = VT. \end{aligned} \quad (\text{A.83})$$

The converse is trickier. We start with the special case where the spectrum of  $T$  is nondegenerate. Let the basis of eigenvectors of  $T$  be labeled  $\mathbf{a}^{(i)}$

$$T\mathbf{a}^{(i)} = \lambda_i \mathbf{a}^{(i)}. \quad (\text{A.84})$$

We assume  $[T, V] = 0$  and we want to prove that  $\mathbf{a}^{(i)}$  is also an eigenvector of  $V$ .

$$[T, V]\mathbf{a}^{(i)} = 0, \quad \text{or} \quad TV\mathbf{a}^{(i)} - VT\mathbf{a}^{(i)} = 0 \quad (\text{A.85})$$



and from Equation A.84

$$\mathbf{T}(\mathbf{V}\mathbf{a}^{(i)}) = \lambda_i (\mathbf{V}\mathbf{a}^{(i)}) . \quad (\text{A.86})$$

Equation A.86 says that the vector  $\mathbf{b}^{(i)} \equiv \mathbf{V}\mathbf{a}^{(i)}$  is an eigenvector of  $\mathbf{T}$  with eigenvalue  $\lambda_i$ . But by assumption, the spectrum of  $\mathbf{T}$  is nondegenerate and that means that  $\mathbf{b}^{(i)}$  must be (up to a constant)  $\mathbf{a}^{(i)}$  itself. If we call the constant  $v_i$ ,

$$\mathbf{b}^{(i)} = \mathbf{V}\mathbf{a}^{(i)} = v_i \mathbf{a}^{(i)} , \quad (\text{A.87})$$

so  $\mathbf{a}^{(i)}$  is an eigenvector of  $\mathbf{V}$ .

All that remains is to relax the assumption of nondegeneracy. Assume now that  $\mathbf{T}$  has at least one degenerate eigenvalue such that both  $\mathbf{a}^{(1)}$  and  $\mathbf{a}^{(2)}$  are eigenvectors of  $\mathbf{T}$  with the same eigenvalue  $\lambda_0$ :

$$\begin{aligned} \mathbf{T}\mathbf{a}^{(1)} &= \lambda_0 \mathbf{a}^{(1)} \\ \mathbf{T}\mathbf{a}^{(2)} &= \lambda_0 \mathbf{a}^{(2)} . \end{aligned}$$

We again assume that the matrices  $\mathbf{T}$  and  $\mathbf{V}$  commute, so

$$\begin{aligned} [\mathbf{T}, \mathbf{V}] \mathbf{a}^{(1)} &= 0 \\ [\mathbf{T}, \mathbf{V}] \mathbf{a}^{(2)} &= 0 , \end{aligned}$$

which leads to the conclusion (as in the nondegenerate case) that both  $\mathbf{b}^{(1)} \equiv \mathbf{V}\mathbf{a}^{(1)}$  and  $\mathbf{b}^{(2)} \equiv \mathbf{V}\mathbf{a}^{(2)}$  are eigenvectors of  $\mathbf{T}$  with eigenvalue  $\lambda_0$ . But this time we can't say that  $\mathbf{b}^{(1)}$  is a constant times  $\mathbf{a}^{(1)}$  since any linear combination of  $\mathbf{a}^{(1)}$  and  $\mathbf{a}^{(2)}$  is an eigenvector of  $\mathbf{T}$  with eigenvalue  $\lambda_0$ . All we know is that

$$\begin{aligned} \mathbf{b}^{(1)} &= \mathbf{V}\mathbf{a}^{(1)} = c_{11} \mathbf{a}^{(1)} + c_{21} \mathbf{a}^{(2)} \\ \mathbf{b}^{(2)} &= \mathbf{V}\mathbf{a}^{(2)} = c_{12} \mathbf{a}^{(1)} + c_{22} \mathbf{a}^{(2)} \end{aligned}$$

for some constants  $c_{ij}$ . So  $\mathbf{a}^{(1)}$  and  $\mathbf{a}^{(2)}$  are not eigenvectors of  $\mathbf{V}$  (unless the constants  $c_{12}$  and  $c_{21}$  just happen to vanish). But suppose we choose a different basis of eigenvectors  $\tilde{\mathbf{a}}$ ,

$$\begin{aligned} \tilde{\mathbf{a}}^{(1)} &= d_{11} \mathbf{a}^{(1)} + d_{21} \mathbf{a}^{(2)} \\ \tilde{\mathbf{a}}^{(2)} &= d_{12} \mathbf{a}^{(1)} + d_{22} \mathbf{a}^{(2)} , \end{aligned} \quad (\text{A.88})$$

for some constants  $d_{ij}$ , such that  $\tilde{\mathbf{a}}^{(1)}$  and  $\tilde{\mathbf{a}}^{(2)}$  are eigenvectors of  $\mathbf{V}$ :

$$\mathbf{V}\tilde{\mathbf{a}}^{(i)} = v_i \tilde{\mathbf{a}}^{(i)} . \quad (\text{A.89})$$

The  $\tilde{\mathbf{a}}$ s are still eigenvectors of  $\mathbf{T}$ , with the same eigenvalue  $\lambda_0$ , since any linear combinations of  $\mathbf{a}^{(1)}$  and  $\mathbf{a}^{(2)}$  are. But *can* we construct linear combinations (A.88) that are eigenvectors of  $\mathbf{V}$ —how do we get the appropriate coefficients  $d_{ij}$ ? *Answer:* We solve the eigenvalue problem<sup>18</sup>

$$\underbrace{\begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}}_{\mathbf{C}} \begin{pmatrix} d_{1i} \\ d_{2i} \end{pmatrix} = v_i \begin{pmatrix} d_{1i} \\ d_{2i} \end{pmatrix} . \quad (\text{A.90})$$

<sup>18</sup> You might worry that the matrix  $\mathbf{C}$  is not diagonalizable, but you need not. The matrix  $\mathbf{C}$  is a  $2 \times 2$  block of the transformation  $\tilde{\mathbf{V}}$  written in the basis  $\mathbf{a}^{(i)}$ ; it is diagonalizable by virtue of the fact that  $\mathbf{V}$  itself is diagonalizable.

I'll let you show (Problem A.24) that the eigenvectors  $\hat{\mathbf{a}}^{(i)}$  constructed in this way satisfy Equation A.88, completing the proof.<sup>19</sup> What we have seen is that, when the spectrum contains degeneracy, the eigenvectors of one matrix aren't *automatically* eigenvectors of a second commuting matrix, but we can always *choose* a linear combination of them to form a simultaneous basis of eigenvectors.

\* **Problem A.18** The  $2 \times 2$  matrix representing a rotation of the  $xy$  plane is

$$\mathbf{T} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (\text{A.91})$$

Show that (except for certain special angles—what are they?) this matrix has no real eigenvalues. (This reflects the geometrical fact that no vector in the plane is carried into itself under such a rotation; contrast rotations in *three* dimensions.) This matrix *does*, however, have *complex* eigenvalues and eigenvectors. Find them. Construct a matrix  $\mathbf{S}$  that diagonalizes  $\mathbf{T}$ . Perform the similarity transformation  $(\mathbf{S}\mathbf{T}\mathbf{S}^{-1})$  explicitly, and show that it reduces  $\mathbf{T}$  to diagonal form.

**Problem A.19** Find the eigenvalues and eigenvectors of the following matrix:

$$\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Can this matrix be diagonalized?

**Problem A.20** Show that the first, second, and last coefficients in the characteristic equation (Equation A.73) are:

$$C_n = (-1)^n, \quad C_{n-1} = (-1)^{n-1} \text{Tr}(\mathbf{T}), \quad \text{and} \quad C_0 = \det(\mathbf{T}). \quad (\text{A.92})$$

For a  $3 \times 3$  matrix with elements  $T_{ij}$ , what is  $C_1$ ?

**Problem A.21** It's obvious that the trace of a *diagonal* matrix is the sum of its eigenvalues, and its determinant is their product (just look at Equation A.79). It follows (from Equations A.65 and A.68) that the same holds for any *diagonalizable* matrix. Prove that in fact

$$\det(\mathbf{T}) = \lambda_1 \lambda_2 \cdots \lambda_n, \quad \text{Tr}(\mathbf{T}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad (\text{A.93})$$

for *any* matrix. (The  $\lambda$ 's are the  $n$  solutions to the characteristic equation—in the case of multiple roots, there may be fewer linearly-independent eigenvectors than there are solutions, but we still count each  $\lambda$  as many times as it occurs.) *Hint*: Write the characteristic equation in the form

$$(\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda) = 0,$$

and use the result of Problem A.20.

<sup>19</sup> I've only proved it for a two-fold degeneracy, but the argument extends in the obvious way to a higher-order degeneracy; you simply need to diagonalize a bigger matrix  $\mathbf{C}$ .

**Problem A.22** Consider the matrix

$$M = \begin{pmatrix} 1 & 1 \\ 1 & i \end{pmatrix}.$$

- (a) Is it normal?
- (b) Is it diagonalizable?

**Problem A.23** Show that if two matrices commute in *one* basis, then they commute in *any* basis. That is:

$$[T_1^e, T_2^e] = 0 \Rightarrow [T_1^f, T_2^f] = 0. \quad (\text{A.94})$$

*Hint:* Use Equation A.64.

**Problem A.24** Show that the  $\tilde{a}$  computed from Equations A.88 and A.90 are eigenvectors of  $V$ .

\*

**Problem A.25** Consider the matrices

$$A = \begin{pmatrix} 1 & 4 & 1 \\ 4 & -2 & 4 \\ 1 & 4 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -2 & -1 \\ -2 & 2 & -2 \\ -1 & -2 & 1 \end{pmatrix}.$$

- (a) Verify that they are diagonalizable and that they commute.
- (b) Find the eigenvalues and eigenvectors of  $A$  and verify that its spectrum is nondegenerate.
- (c) Show that the eigenvectors of  $A$  are eigenvectors of  $B$  as well.

**Problem A.26** Consider the matrices

$$A = \begin{pmatrix} 2 & 2 & -1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 5 & -1 \\ 2 & -1 & 2 \end{pmatrix}.$$

- (a) Verify that they are diagonalizable and that they commute.
- (b) Find the eigenvalues and eigenvectors of  $A$  and verify that its spectrum is degenerate.
- (c) Are the eigenvectors that you found in part (b) also eigenvectors of  $B$ ? If not, find the vectors that are simultaneous eigenvectors of both matrices.

## A.6 HERMITIAN TRANSFORMATIONS

In Equation A.48 I defined the hermitian conjugate (or “adjoint”) of a *matrix* as its transpose-conjugate:  $T^\dagger = \tilde{T}^*$ . Now I want to give you a more fundamental definition for the hermitian conjugate of a *linear transformation*: It is that transformation  $\hat{T}^\dagger$  which, when applied to the

first member of an inner product, gives the same result as if  $\hat{T}$  itself had been applied to the second vector:

$$\langle \hat{T}^\dagger \alpha | \beta \rangle = \langle \alpha | \hat{T} \beta \rangle, \quad (\text{A.95})$$

(for all vectors  $|\alpha\rangle$  and  $|\beta\rangle$ ).<sup>20</sup> I have to warn you that although everybody uses it, this is lousy notation. For  $\alpha$  and  $\beta$  are not *vectors* (the *vectors* are  $|\alpha\rangle$  and  $|\beta\rangle$ ), they are *names*. In particular, they are endowed with no mathematical properties at all, and the expression “ $\hat{T}\beta$ ” is literally *nonsense*: Linear transformations act on *vectors*, not *labels*. But it’s pretty clear what the notation *means*:  $\hat{T}\beta$  is the name of the vector  $\hat{T}|\beta\rangle$ , and  $\langle \hat{T}^\dagger \alpha | \beta \rangle$  is the inner product of the vector  $\hat{T}^\dagger |\alpha\rangle$  with the vector  $|\beta\rangle$ . Notice in particular that

$$\langle \alpha | c\beta \rangle = c \langle \alpha | \beta \rangle, \quad (\text{A.96})$$

whereas

$$\langle c\alpha | \beta \rangle = c^* \langle \alpha | \beta \rangle, \quad (\text{A.97})$$

for any scalar  $c$ .

If you’re working in an orthonormal basis (as we always do), the hermitian conjugate of a linear transformation is represented by the hermitian conjugate of the corresponding matrix; for (using Equations A.50 and A.53),

$$\langle \alpha | \hat{T} \beta \rangle = \mathbf{a}^\dagger \mathbf{T} \mathbf{b} = (\mathbf{T}^\dagger \mathbf{a})^\dagger \mathbf{b} = \langle \hat{T}^\dagger \alpha | \beta \rangle. \quad (\text{A.98})$$

So the terminology is consistent, and we can speak interchangeably in the language of transformations or of matrices.

In quantum mechanics, a fundamental role is played by **hermitian transformations** ( $\hat{T}^\dagger = \hat{T}$ ). The eigenvectors and eigenvalues of a hermitian transformation have three crucial properties:

### 1. The eigenvalues of a hermitian transformation are real.

**Proof:** Let  $\lambda$  be an eigenvalue of  $\hat{T}$ :  $\hat{T}|\alpha\rangle = \lambda|\alpha\rangle$ , with  $|\alpha\rangle \neq |0\rangle$ . Then

$$\langle \alpha | \hat{T} \alpha \rangle = \langle \alpha | \lambda \alpha \rangle = \lambda \langle \alpha | \alpha \rangle.$$

Meanwhile, if  $\hat{T}$  is hermitian, then

$$\langle \alpha | \hat{T} \alpha \rangle = \langle \hat{T} \alpha | \alpha \rangle = \langle \lambda \alpha | \alpha \rangle = \lambda^* \langle \alpha | \alpha \rangle.$$

But  $\langle \alpha | \alpha \rangle \neq 0$  (Equation A.20), so  $\lambda = \lambda^*$ , and hence  $\lambda$  is real. QED

### 2. The eigenvectors of a hermitian transformation belonging to distinct eigenvalues are orthogonal.

**Proof:** Suppose  $\hat{T}|\alpha\rangle = \lambda|\alpha\rangle$  and  $\hat{T}|\beta\rangle = \mu|\beta\rangle$ , with  $\lambda \neq \mu$ . Then

$$\langle \alpha | \hat{T} \beta \rangle = \langle \alpha | \mu \beta \rangle = \mu \langle \alpha | \beta \rangle,$$

<sup>20</sup> If you’re wondering whether such a transformation necessarily *exists*, that’s a good question, and the answer is “yes.” See, for instance, Paul R. Halmos, *Finite Dimensional Vector Spaces*, 2nd edn, van Nostrand, Princeton (1958), Section 44.

and if  $\hat{T}$  is hermitian,

$$\langle \alpha | \hat{T} \beta \rangle = \langle \hat{T} \alpha | \beta \rangle = \langle \lambda \alpha | \beta \rangle = \lambda^* \langle \alpha | \beta \rangle.$$

But  $\lambda = \lambda^*$  (from (1)), and  $\lambda \neq \mu$ , by assumption, so  $\langle \alpha | \beta \rangle = 0$ . QED

### 3. The eigenvectors of a hermitian transformation span the space.

As we have seen, this is equivalent to the statement that any hermitian matrix can be diagonalized. This rather technical fact is, in a sense, the mathematical support on which much of quantum mechanics leans. It turns out to be a thinner reed than one might have hoped, because the proof does not carry over to infinite-dimensional vector spaces.

**Problem A.27** A hermitian linear transformation must satisfy  $\langle \alpha | \hat{T} \beta \rangle = \langle \hat{T} \alpha | \beta \rangle$  for all vectors  $|\alpha\rangle$  and  $|\beta\rangle$ . Prove that it is (surprisingly) sufficient that  $\langle \gamma | \hat{T} \gamma \rangle = \langle \hat{T} \gamma | \gamma \rangle$  for all vectors  $|\gamma\rangle$ . *Hint:* First let  $|\gamma\rangle = |\alpha\rangle + |\beta\rangle$ , and then let  $|\gamma\rangle = |\alpha\rangle + i|\beta\rangle$ .

\*

**Problem A.28** Let

$$T = \begin{pmatrix} 1 & 1-i \\ 1+i & 0 \end{pmatrix}.$$

- Verify that  $T$  is hermitian.
- Find its eigenvalues (note that they are real).
- Find and normalize the eigenvectors (note that they are orthogonal).
- Construct the unitary diagonalizing matrix  $S$ , and check explicitly that it diagonalizes  $T$ .
- Check that  $\det(T)$  and  $\text{Tr}(T)$  are the same for  $T$  as they are for its diagonalized form.

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**Problem A.29** Consider the following hermitian matrix:

$$T = \begin{pmatrix} 2 & i & 1 \\ -i & 2 & i \\ 1 & -i & 2 \end{pmatrix}.$$

- Calculate  $\det(T)$  and  $\text{Tr}(T)$ .
- Find the eigenvalues of  $T$ . Check that their sum and product are consistent with (a), in the sense of Equation A.93. Write down the diagonalized version of  $T$ .
- Find the eigenvectors of  $T$ . Within the degenerate sector, construct two linearly independent eigenvectors (it is this step that is always possible for a *hermitian* matrix, but not for an *arbitrary* matrix—contrast Problem A.19). Orthogonalize them, and check that both are orthogonal to the third. Normalize all three eigenvectors.
- Construct the unitary matrix  $S$  that diagonalizes  $T$ , and show explicitly that the similarity transformation using  $S$  reduces  $T$  to the appropriate diagonal form.

**Problem A.30** A **unitary transformation** is one for which  $\hat{U}^\dagger \hat{U} = 1$ .

- Show that unitary transformations preserve inner products, in the sense that  $\langle \hat{U}\alpha | \hat{U}\beta \rangle = \langle \alpha | \beta \rangle$ , for all vectors  $|\alpha\rangle, |\beta\rangle$ .

- (b) Show that the eigenvalues of a unitary transformation have modulus 1.  
 (c) Show that the eigenvectors of a unitary transformation belonging to distinct eigenvalues are orthogonal.

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**Problem A.31** Functions of matrices are typically defined by their Taylor series expansions. For example,

$$e^{\mathbf{M}} \equiv \mathbf{I} + \mathbf{M} + \frac{1}{2}\mathbf{M}^2 + \frac{1}{3!}\mathbf{M}^3 + \cdots \quad (\text{A.99})$$

- (a) Find  $\exp(\mathbf{M})$ , if

$$(i) \mathbf{M} = \begin{pmatrix} 0 & 1 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix}; \quad (ii) \mathbf{M} = \begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix}.$$

- (b) Show that if  $\mathbf{M}$  is diagonalizable, then

$$\det(e^{\mathbf{M}}) = e^{\text{Tr}(\mathbf{M})}. \quad (\text{A.100})$$

*Comment:* This is actually true even if  $\mathbf{M}$  is *not* diagonalizable, but it's harder to prove in the general case.

- (c) Show that if the matrices  $\mathbf{M}$  and  $\mathbf{N}$  commute, then

$$e^{\mathbf{M}+\mathbf{N}} = e^{\mathbf{M}}e^{\mathbf{N}}. \quad (\text{A.101})$$

Prove (with the simplest counterexample you can think up) that Equation A.101 is *not* true, in general, for *non*-commuting matrices.<sup>21</sup>

- (d) If  $\mathbf{H}$  is hermitian, show that  $e^{i\mathbf{H}}$  is unitary.

<sup>21</sup> See Problem 3.29 for the more general “Baker–Campbell–Hausdorff” formula.